

Effect of hormonal changes during menstrual cycle on women mental wellbeing

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Introduction/Background

Fluctuations in estradiol and progesterone throughout the menstrual cycle are closely linked to variations in women's mood and mental wellbeing. Physical and psychological symptoms tend to intensify during the late luteal and early follicular phases, when hormone levels decline. These cyclical changes may influence mood, anxiety, and behavior across the different phases of the cycle.

Objective

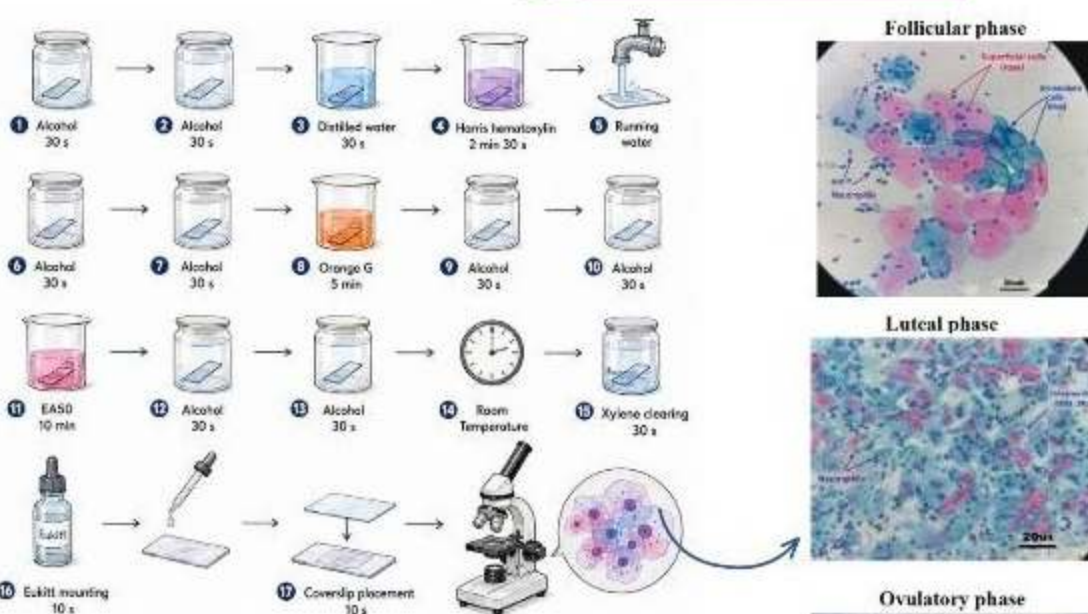
To investigate the relationship between hormonal fluctuations, women's mental wellbeing, and behavioral variations across the menstrual cycle phases.

Material and Methods

Vaginal cytology was used to identify the cycle phases. Plasma levels of luteinizing hormone, follicle-stimulating hormone, estradiol and progesterone were measured across the different phases: menstruation, follicular phase, ovulation, and early luteal phase from the participants of Nafissa Hamoud Hospital (Algiers). The participants had an integrated two steps procedure comprising Papanicolaou smear sample collection and hormonal analysis performed during their menstrual cycle phases.

Results

Papanicolaou staining



Hormonal result

Table. Hormonal profile during different phases of the menstrual cycle

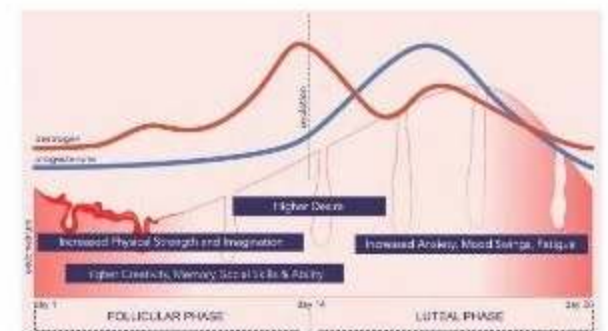
Cycle phase	Hormone	Estradiol (pg/mL)	Progesterone (ng/mL)	LH (mIU/ml)	FSH (mIU/ml)
Menstruation (J1-J5)		30-50	<1	10-15	10-15
Follicular phase (J6-J13)		50-150	<1	15-25	8-12
Ovulation (~J14)		180-200	1-2	55-60	25-30
Early luteal phase (J15-J21)		80-120	8-10	20-25	10-12

Plasma levels of luteinizing hormone, follicle-stimulating hormone, estradiol and progesterone were measured across the different phases of the menstrual cycle: menstruation, follicular phase, ovulation, and early luteal phase from the patients of Nafissa Hamoud Hospital (Algiers).

Women menstrual cycle

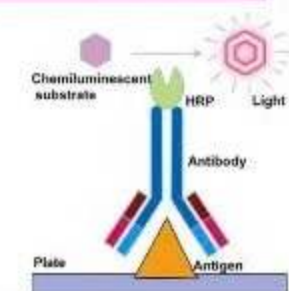


Mood behavior during women menstrual cycle



Estrogen, in particular, plays an important role in supporting mood-regulating chemicals, when levels fluctuate it can lead to emotional changes.

ECLIA hormone assay



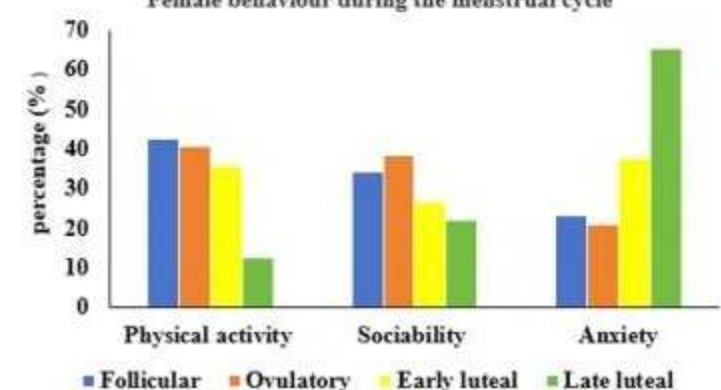
Survey : Mood and menstrual cycle



Questionnaire sur les variations comportementales durant le cycle menstruel
استبيان حول التغيرات السلوكية خلال الدورة الشهرية
Veuillez répondre selon votre ressenti personnel durant chaque phase du cycle menstruel.
يرجى الإجابة حسب شعورك الشخصي خلال كل مرحلة من مراحل الدورة الشهرية.

Survey

Female behaviour during the menstrual cycle



Our questionnaire data collected from 63 women of reproductive age revealed a clear and consistent behavioural pattern variation across the menstrual cycle. Physical activity and sociability peaked during the follicular and ovulatory phases coinciding with rising and peak estradiol levels. Both parameters declined sharply in the late luteal phase, while irritability followed an inverse trajectory, surging to increase relative to the ovulatory phase. This behavioural profile is consistent with the clinical criteria for premenstrual syndrome, with irritability reported as the most common of late luteal symptom.

Conclusion

Moreover, these findings underscore the importance of the endocrine system in the regulation of anxiety in women, and strongly support the routine consideration of monitoring and tracking the menstrual cycle as a variable of interest when studying anxiety. Further research is needed to validate the utility and precision of hormone use in the field of mental health.